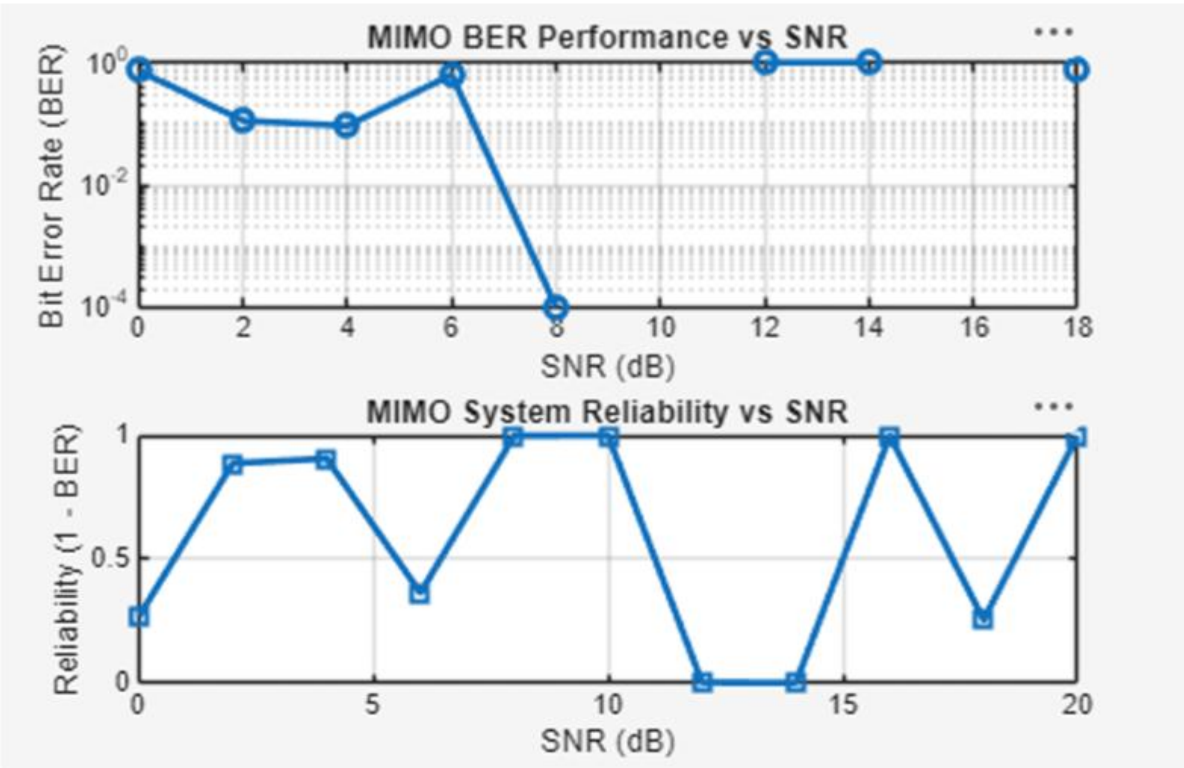
EXPERIMENT-05

OBJECTIVE-5.1

```
clc;

clear;

close all;

% SNR range

SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters

Nt = 2;

Nr = 2;

BER = zeros(1,length(SNR));

clc;

clear;

close all;

% SNR range
```

```
SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters

Nt = 2;

Nr = 2;

BER = zeros(1,length(SNR));
```

OUTPUT:-



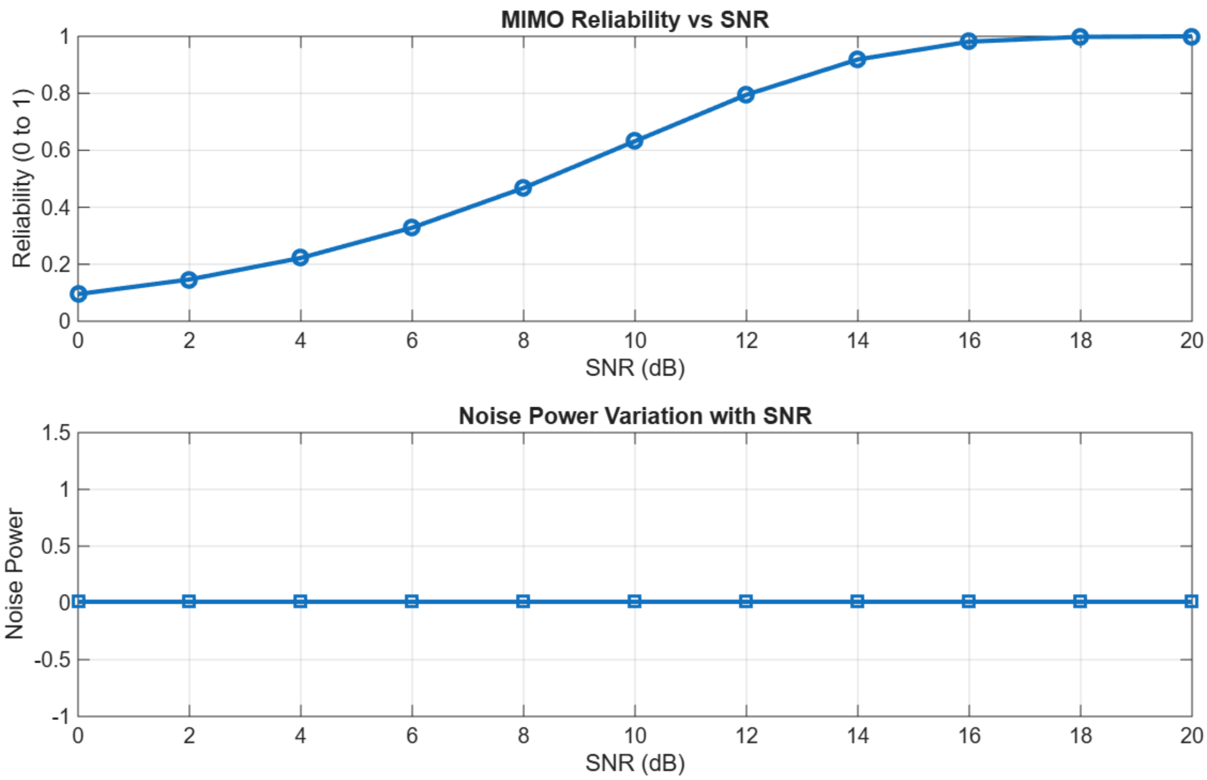
OBJECTIVE-5.2

```
clc;

clear;

close all;

% SNR range (dB)

SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters

Nt = 2;

Nr = 2;

% Simulated reliability metric (inverse BER model)

Reliability = zeros(1,length(SNR));

for i = 1:length(SNR)

    % simple channel model effect

    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
```

```
    noise_power = 1/SNR(i);

    % reliability increases with SNR

    Reliability(i) = 1 - exp(-SNR(i)/10);

end

figure

% -------- Graph 1: SNR vs Reliability --------

subplot(2,1,1)

plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)

title('MIMO Reliability vs SNR')

xlabel('SNR (dB)')

ylabel('Reliability (0 to 1)')

grid on

% -------- Graph 2: SNR vs Noise Effect --------

subplot(2,1,2)
```

```
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)

title('Noise Power Variation with SNR')

xlabel('SNR (dB)')

ylabel('Noise Power')

grid on
```

OUTPUT:-



OBJECTIVE-5.3

```
clc;

clear;

close all;

% Resource Block parameters

rb_bw = 180e3;   % 180 kHz per RB

% Users

users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [10 12 8 15 5];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2];   % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

% Display results
```

```
disp('User Throughput (Mbps):')

disp(throughput')

figure

% -------- Graph 1: Throughput per User --------

subplot(2,1,1)

plot(users, throughput, '-o','LineWidth',2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2: RB vs Throughput --------

subplot(2,1,2)

plot(alloc_RB, throughput, '-s','LineWidth',2)

title('Resource Blocks vs Throughput')
```

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on

OUTPUT:-